MIL-F-45569B(AL)
31 May 1983
SUPERSEDING
MIL-B-45569A(AL)
29 June 1978

MILITARY SPECIFICATION

FIXTURE, LATHE, METAL CUTTING

This specification is approved for use by the US Army Armament Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of The Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a lathe fixture for holding a workpiece while performing milling, drilling, slotting, tapping, layout, boring, and indexing operations. It is adapted for compound mounting on 10-inch lathes.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-1150 - Tools and Tool Accessories for Power-Driven Metal and Woodworking Machinery, Packaging and Packing of

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards For Federal Services

Beneficial comments (recommendations, additions; deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Materiel Readiness Command, ATTN: DRSAR-LEE-S, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3465

Source: https://assist.dla.mil -- Downloaded: 202610-22T03:25Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one complete fixture for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The fixture shall be new and of the current standard commercial meeting the performance requirements of this specification. All parts subject to wear or replacement shall be accessible for adjustment or repair. The fixture shall provide a means of holding round and rectangular workpieces, and shall include a graduated vertical feed. The fixture shall be adaptable for mounting on the compound of the lathe.

3.2.1 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible. Lubrication reservoirs, if applicable, shall have means for checking levels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

3.2.2 Threads. All threaded parts of the fixture shall conform to FED-STD-H28, and the applicable Detailed Standard section referenced therein.

3.2.3 Interchangeability. All replaceable parts shall be designed and manufactured according to standards that shall permit replacement without the need for modification of the parts or the fixture.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry. When dissimilar metals are in contact with each other, other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissable to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction. The fixture shall be furnished completely so that, when installed, it can be used for all the operations for which it was designed. Only that reclamation of defective parts or assemblies which retains original strength, performance, interchangeability, appearance, and dimensions shall be allowed.

3.4.1 Castings and forgings. Castings shall be free of blowholes, porosity, hard spots, shrinkage defects, cracks, and other defects. Forgings shall be free of scale, inclosures, cold shuts, mismatching, and other defects that could affect structural strength. The physical and chemical properties of castings and forgings shall be adequate to meet the performance requirements of this specification.

3.4.2 Sizes and capacities. The fixture shall be a size that is adaptable for use on ten-inch lathes. The fixture shall have a vertical adjustment of not less than 3 1/2 inches. The working surface (fixture plate) shall not be less than 5 by 5 inches. The fixture shall be capable of holding round stock up to two inches in diameter, and square stock up to two inches.

3.5 Performance and accuracy characteristics.

3.5.1 Fixture. The fixture shall be secured to the lathe without distortion of the fixture or the lathe. Vertical adjustment shall be accomplished in increments of .001 inch. An error in adjustment shall not exceed 0.0005 of an inch per inch of adjustment. The working surface (fixture plate) shall be hardened to prevent wear and shall be finish ground.

3.5.1.1 Key seat cutting. The fixture shall hold round bar stock for key seat cutting. The surfaces of the cuts shall be free of chatter marks.

3.5.1.2 Keyway cutting. The fixture shall securely hold round bar stock while cutting keyways in one continuous operation. Keyway cuts shall not vary more than .003 inch per foot from a straight line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

3.5.1.3 Vertical slotting. The fixture shall be capable of holding the workpiece while milling a vertical slot. Milling shall be accomplished by using a vertical feed on the fixture and an end mill in the lathe. The accuracy of a predetermined slot width shall be ±.001 inch. The slot shall be perpendicular to the horizontal and shall have an accuracy of ±.0015 per inch of slot length.

3.5.2 Indexing device. The indexing device shall be graduated in degrees and shall accommodate round bar stocks from one-fourth inch in diameter to two inches in diameter. The device shall be capable of being secured in any position from 0 through 360 degrees with an accuracy of ±30 minutes when measured from the base cut.

3.5.3 Rigidity. The rigidity of the fixture shall produce a 125 micro-inch arithmetical average (AA) maximum finish on a one inch square bar of AISI 1045 cold rolled steel after a cut of one-half inch depth, full width, for a distance of 2 1/2 inches, at a rate of 1 3/4 inches per minute (minimum) using a one inch diameter end mill.

3.6 Standard equipment and accessories. The fixture shall be furnished with the necessary standard equipment (see 6.2) to permit mounting on lathes, including all the necessary work holding clamps and fasteners needed to perform the operations listed herein.

3.6.1 Case. A metal case of sufficient size shall be furnished. This case shall include brackets, clamps, and any other holding devices needed to store and secure the fixture, instructions, spare parts, and wrenches. The contents shall be arranged/secured within the case in such a manner that they shall not be displaced during normal handling. The case shall be designed to provide easy access to the contents.

3.7 Painting. Unless otherwise specified (see 6.2), all painting and finishing of the fixture and case shall be in accordance with standard commercial practice provided the following criteria are met or exceeded: all surfaces to be painted shall be clean and free of any foreign matter detrimental to painting, and shall have at least one coat of primer and one coat of enamel, or equivalent, applied.

3.8 Product Identification. Nameplate(s) permanently and legibly marked in accordance with MIL-STD-130, including serial number, shall be securely attached to each fixture and, unless otherwise specified (see 6.2), shall include the National Stock Number.

3.9 Workmanship. Standards of workmanship shall assure that the lathe fixture shall have the efficient operating characteristics found in standard commercial units and as specified in Section 3 herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Tests (see 4.6).
(c) Packaging inspection (see 4.7).

4.4.1 Sampling. When required (see 6.2), sampling for quality conformance inspection shall be as specified in MIL-STD-105 at the Inspection Level and Acceptable Quality Level (AQL) specified in the contract. For packaging inspection (see 4.7), special inspection level S-4 and an AQL of 4.0 percent defective shall be used. General inspection level I, with no failures acceptable, shall be used for tests.

4.5 Product examination. Each complete lathe fixture shall be examined to determine compliance with the requirements in Section 3. As a minimum, the following characteristics shall be examined:

CHARACTERISTICS	REQUIREMENT PARAGRAPH
Design not as specified	3.2
Lubrication not as specified	3.2.1
Threads not as specified	3.2.2
Interchangeability not as specified	3.2.3
Materials not as specified	3.3
Construction not as specified	3.4
Defective castings or forgings	3.4.1
Sizes or capacities not as specified	3.4.2
Equipment not as specified	3.6
Painting not as specified	3.7
Identification marking not as specified	3.8
Workmanship not as specified	3.9

4.6 Tests. Each fixture shall be tested to determine compliance with performance requirements as specified in 4.6.1 through 4.6.5.

Source: https://assist.dla.mil -- Downloaded: 202610-22T03:25Z
Check the source to verify that this is the current version before use.

4.6.1 Key seat cutting test. The fixture shall be mounted on a lathe. Using a No. 204 Woodruff cutter (or equal) and a No. 608 Woodruff cutter (or equal), respectively, one key seat shall be cut in a three-quarter inch diameter bar and in a two inch diameter bar of AISI 1045 steel. Key seats shall be examined for compliance with 3.5.1.1.

4.6.2 Keyway cutting test. The fixture shall be mounted on a lathe. Using a minimum feed of 1 3/4 inches per minute, one keyway measuring 1/4 inch deep x 1/2 inch wide x 6 inches long shall be cut in a two inch round bar of AISI 1045 steel. The keyway shall be examined/tested for compliance with 3.5.1.2.

4.6.3 Vertical slotting test. The fixture shall be mounted on a lathe. One vertical slot measuring .125 inches deep, .375 inches wide, and one inch long shall be cut in a 1/4 inch deep by 1 1/2 inch wide by 6 inches long piece of AISI 1045 steel. Accuracy shall be as specified in 3.5.1.3.

4.6.4 Rigidity. The basic fixture shall be mounted on a machine of sufficient size, rigidity, and horsepower to conduct this test. A test specimen, a one inch square bar of AISI 1045 cold rolled steel, shall be mounted on the fixture. Using a one inch diameter end mill, a cut shall be taken, as specified in 3.5.3. The test specimen shall then be examined for compliance with 3.5.3.

4.6.5 Indexing test. The indexing device and basic fixture shall be mounted on a lathe, and test cuts shall be made on the periphery of a two inch diameter bar made of AISI 1045 cold rolled steel. Cuts shall be made one-eighth inch wide, one-eighth inch deep, and one inch long. Cuts shall be made at 45 degrees, 90 degrees, 135 degrees, 180 degrees, and 270 degrees from the first cut. The cuts shall be examined for compliance with 3.5.2.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with Section 5 requirements.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be Level A in accordance with PPP-T-1150, or Industrial in accordance with ASTM D3951-82, as specified (see 6.2).

6. NOTES

6.1 Intended use. The lathe fixture covered by this specification is intended for use in milling, drilling, slotting, tapping, layout, boring, turning, and indexing operations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article inspection when required (see 3.1).
(c) Standard equipment and accessories if different (see 3.6).
(d) Painting if different (see 3.7).
(e) Identification marking if different (see 3.8).
(f) First article inspection if different (see 4.3).
(g) Quality conformance inspection if different (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.

(h) Sampling when required (see 4.4.1).
(i) Selection of applicable level of preservation, packaging, packing, and marking required (see 5.1).

6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and controls of noise levels, radiation, electromagnetic emissions, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.4 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - AL

Preparing activity:

Army - AL

Review activities:

Army - SM

Project No. 3465-A003

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:25Z
Check the source to verify that this is the current version before use.